Ender Batu Demirtaş

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Education

California State University, Fullerton — Fullerton, CA
Bachelor of Science in Computer Science — *May 2026*

- **Relevant Coursework:** Data Structures, Software Engineering, Operating Systems, OOP, Back-end Development, Machine Learning

Experience

Catalyst Insight Studio – General Chemistry II — Fullerton, CA
Co-Developer — *May 2026 – July 2026*

- Enabled General Chemistry II students and instructors to explore lab results in the browser with no install, by co-building a **Streamlit** app and owning its **pandas** data pipeline and **scikit-learn** models.
- Turned messy student survey exports into analysis-ready data, by building the normalization, cleaning, and validation pipeline.
- Kept the app working as the lab's data-collection forms evolved, by adding automated schema-mismatch validation and a **New Semester** tab for merging raw Excel exports.
- Eased handoff to the next student team, by breaking up a monolithic app into separate UI and data layers and writing **54 unit tests**.

Association for Computing Machinery (ACM) – CSUF — Fullerton, CA
Artificial Intelligence Board Member — *January 2025 – May 2026*

- Led an overhaul of presentation and workshop formats, increasing attendance by **30%+**.
- Revamped our engagement strategies, growing activity participation from negligible to **25+ students** in interactive activities.
- Contributed to a more dynamic and structured AI and ML learning environment, which is both accessible to beginners while also providing depth for experienced members and learners.

Data Science & Machine Learning Club (DSML) – CSUF — Fullerton, CA
Vice President — *January 2025 – May 2026*

- Lead weekly presentations on core data science and machine learning concepts, combining theory with real-world applications.
- Implement live tool demonstrations to showcase practical application of core data science concepts.
- Host hybrid sessions to maximize accessibility, maintaining high attendance across both online and in-person audiences.
- Design case study-driven curricula using realistic datasets to mirror production machine learning challenges.

Projects

NL2SQL Analytics Platform | *Python, FastAPI, Strands Agents, AWS* — September 2025 – November 2025

- Developed an agentic AI chatbot that democratizes data access, enabling non-technical users to query structured PostgreSQL databases using natural language.
- Orchestrated AI reasoning and dynamic tool execution via **Strands Agents** and **Amazon Bedrock** (Claude 3.5), translating plain English into precise, case-insensitive SQL queries while mitigating hallucinations.
- Engineered a robust REST API using **FastAPI** and **Pydantic**, securely connecting to AWS RDS by dynamically retrieving credentials from **AWS Secrets Manager** to ensure security.
- Deployed a serverless, containerized backend to **AWS App Runner** leveraging the Amazon Bedrock AgentCore runtime, establishing an automated CI/CD deployment pipeline directly from GitHub.

Technical Skills

Programming Languages: Python, C++, Swift, SQL
Cloud: AWS (EC2, S3, IAM, Bedrock, KMS, EBS)
Frameworks & Libraries: FastAPI, SwiftUI, SwiftData, LangGraph
Tools & Collaboration: Git, GitHub Kanban, Jira, Confluence, Slack
Technical Concepts: Algorithms, Machine Learning, RAG, Agentic AI, APIs

Leadership & Social Influence

Technical Event Production & Outreach — Fullerton, CA
Bridging the Gap Between Students and Industry — *January 2025 – December 2025*

- Founded and coordinated a guest speaker series, hosting engineering leaders from companies including AWS to deliver panels on production architectures and data engineering.
- Attracted over **20+ students per session** by promoting interactive discussions on modern cloud and back-end practices.
- Facilitated industry networking opportunities to prepare student members for technical interviews and post-grad career pathways.